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Momentum & Collisions Quizz



Summary

2025

Question	Average Grade	Standard Deviation	Discrimination Index	Point Biserial
Question 1	54.84 %	49.77 %	87.50 %	0.64
Question 2	87.1 %	33.52 %	50.00 %	0.54
Question 3	51.61 %	49.97 %	87.50 %	n/a
Question 4	22.58 %	41.81 %	50.00 %	n/a
Question 5	67.74 %	46.75 %	62.50 %	0.50
Question 6	64.52 %	47.85 %	62.50 %	0.43
Question 7	51.61 %	49.97 %	62.50 %	0.36

2026

Question	Average Grade	Standard Deviation	Discrimination Index	Point Biserial
Question 1	80 %	40.00 %	50.00 %	0.42
Question 2	86.67 %	33.99 %	0.00 %	-0.02
Question 3	40 %	48.99 %	100.00 %	n/a
Question 4	53.33 %	49.89 %	75.00 %	n/a
Question 5	73.33 %	44.22 %	75.00 %	0.76
Question 6	86.67 %	33.99 %	25.00 %	0.50
Question 7	73.33 %	44.22 %	50.00 %	0.37

Questions 3 and 4

3

An open train car runs on a straight horizontal track with constant speed (disregard air resistance and other losses). It starts to rain and water accumulates in the car. We consider the system comprising the train car and the accumulated water.

4

An open train car runs on a straight horizontal track with constant speed (disregard air resistance and other losses). The train car contains some water at the bottom. A valve in the bottom is opened so the water is spilling out.

What happens to the speed of the system comprising train car with water?

1. Increasing.

What happens to the momentum of the system comprising train car with water?

2. Decreasing.

What happens to the kinetic energy of the system comprising train car with water?

3. No change.



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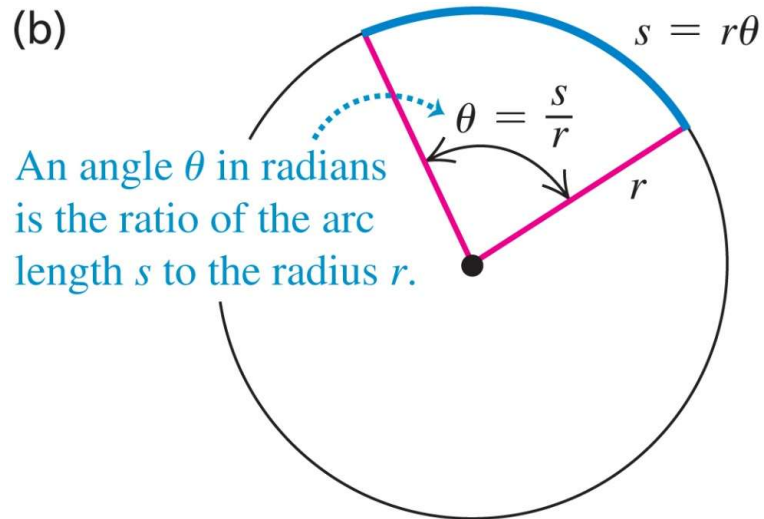
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Recap on Rotation

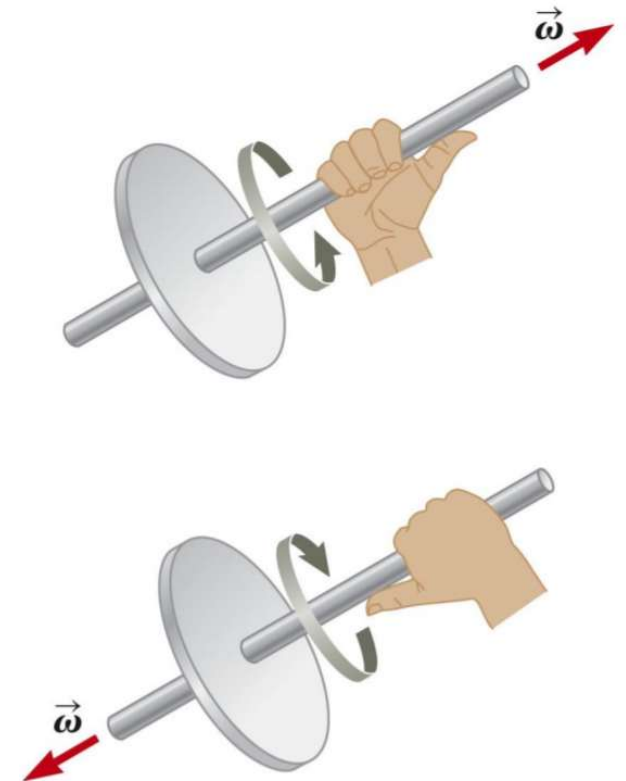
Definition of angle, angular velocity etc.



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$$\omega_z = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t} = \frac{d\theta}{dt}$$

$$\alpha_z = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega_z}{\Delta t} = \frac{d\omega_z}{dt}$$



Summary of Linear-Rotary comparison

Straight-Line Motion with Constant Linear Acceleration

$$a_x = \text{constant}$$

$$v_x = v_{0x} + a_x t$$

$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$$

Fixed-Axis Rotation with Constant Angular Acceleration

$$\alpha_z = \text{constant}$$

$$\omega_z = \omega_{0z} + \alpha_z t$$

$$\theta = \theta_0 + \omega_{0z} t + \frac{1}{2} \alpha_z t^2$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t$$

$$x \rightarrow \theta$$

$$v_x \rightarrow \omega$$

$$a_x \rightarrow \alpha$$

$$m \rightarrow I$$

Kinetic energy of a particle $\rightarrow K = \frac{1}{2}mv^2$ $\leftarrow$ **Mass of particle**
 $\leftarrow$ **Speed of particle**

Rotational kinetic energy of a rigid body rotating around an axis $\rightarrow K = \frac{1}{2}I\omega^2$ $\leftarrow$ **Moment of inertia of body for given rotation axis**
 $\leftarrow$ **Angular speed of body**

Kinetic Energy

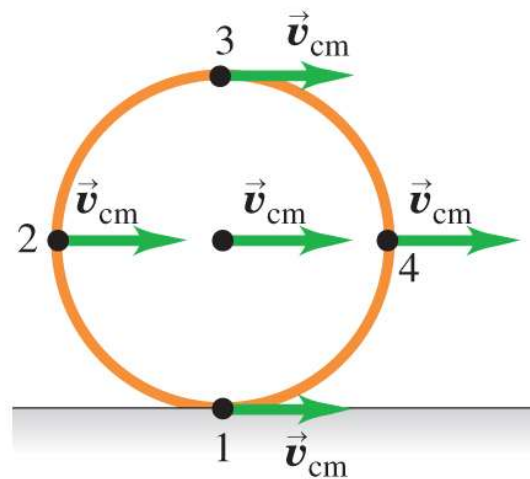
Kinetic energy of a rigid body with both translation and rotation

$$K = \frac{1}{2} M v_{\text{cm}}^2 + \frac{1}{2} I_{\text{cm}} \omega^2$$

Kinetic energy of *translation* of center of mass (cm)
 Kinetic energy of *rotation* around axis through cm
 Mass of rigid body
 Speed of cm
 Moment of inertia of rigid body about axis through cm
 Angular speed of rigid body

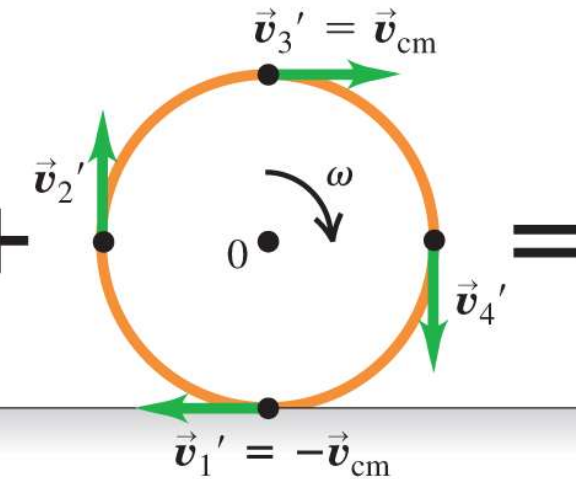
Rolling without slipping: Translation + Rotation

Translation of center of mass:
velocity $\vec{v}_{cm}$

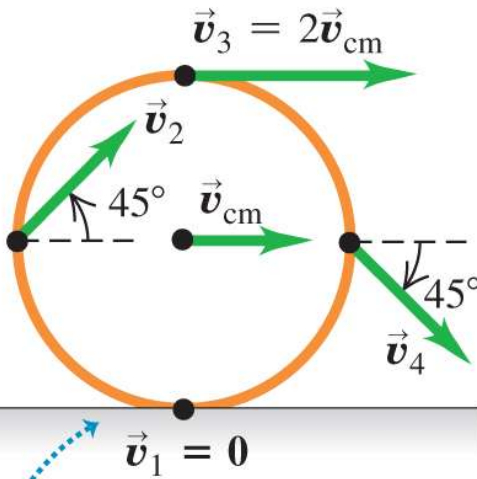


$$v_{CM} = R \cdot \omega$$

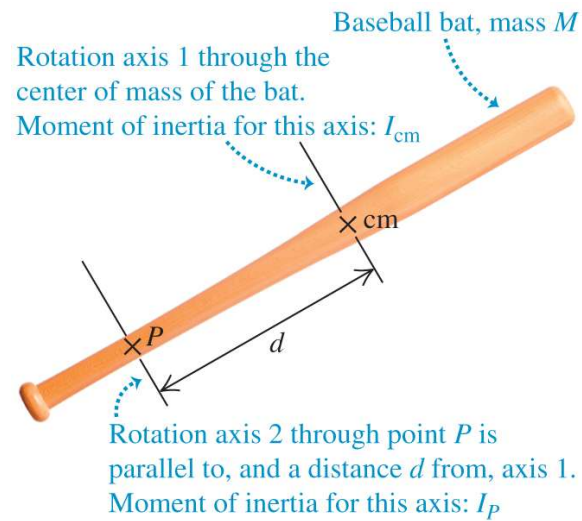
Rotation around center of mass:
for rolling without slipping,
speed at rim = v_{cm}



Combined motion



Parallel-axis (Steiner) theorem



Parallel-axis theorem: $I_P = I_{cm} + Md^2$

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Parallel-axis theorem:

Moment of inertia of a body for a rotation axis through point P

$$I_P = I_{cm} + Md^2$$

Moment of inertia of body for a parallel axis through center of mass

Mass of body

Distance between two parallel axes

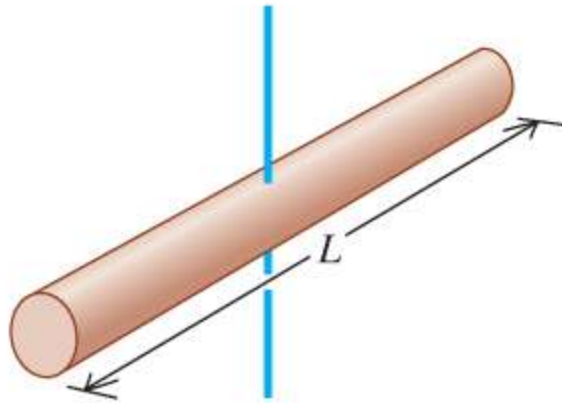
(9.19)

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Parallel-axis (Steiner) theorem for a rod

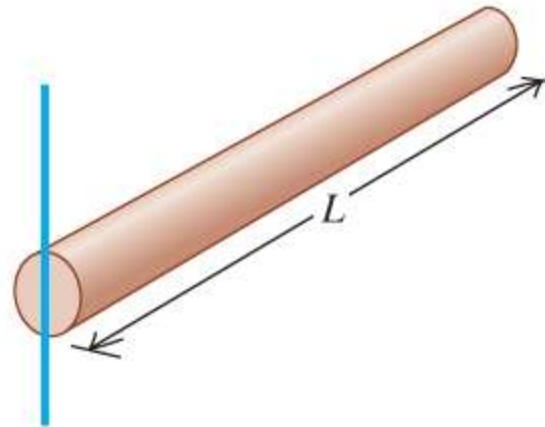
(a) Slender rod,
axis through center

$$I = \frac{1}{12}ML^2$$



(b) Slender rod,
axis through one end

$$I = \frac{1}{3}ML^2$$

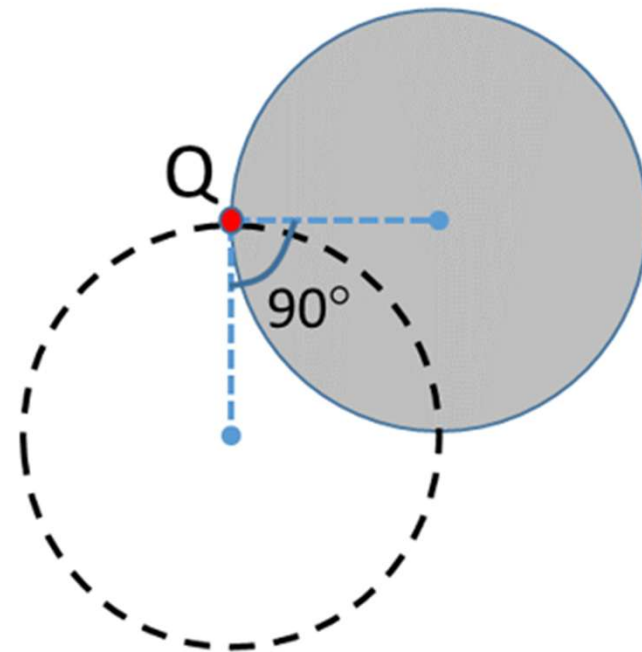


Problem from the exam: Using Steiner Theorem

A uniform solid disk with radius $R = 30 \text{ cm}$ and mass $M = 1 \text{ kg}$. can freely rotate through an axis Q that passes through its edge (see Figure). Initially the disk is kept at rest in the position shown by the grey circle in the figure. Then the disk is freed and starts moving.

Data: Moment of inertia of a disk through an axis passing through its center of mass $I_{CM} = \frac{1}{2} M R^2$. Use $g = 10 \text{ m/s}^2$.

What is the speed of the center of mass of the disk when it reaches the position indicated by the dashed line circle?



Acceleration in rotating bodies

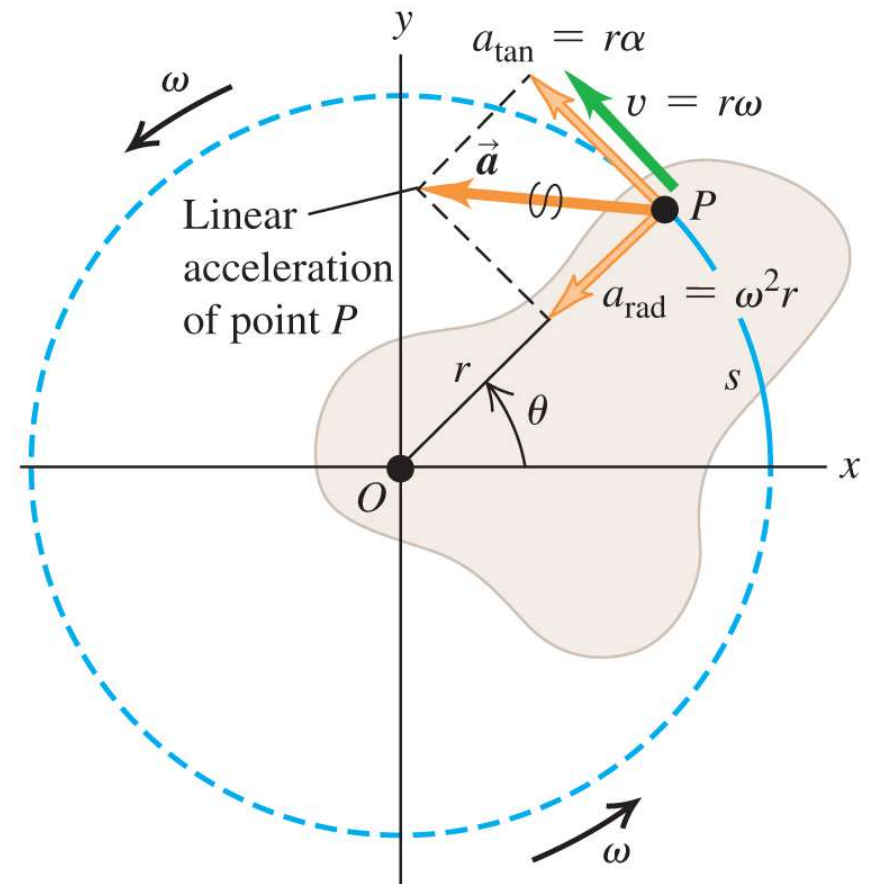
Tangential acceleration of a point on a rotating rigid body

Distance of that point from rotation axis

$$a_{\text{tan}} = \frac{dv}{dt} = r \frac{d\omega}{dt} = r\alpha$$

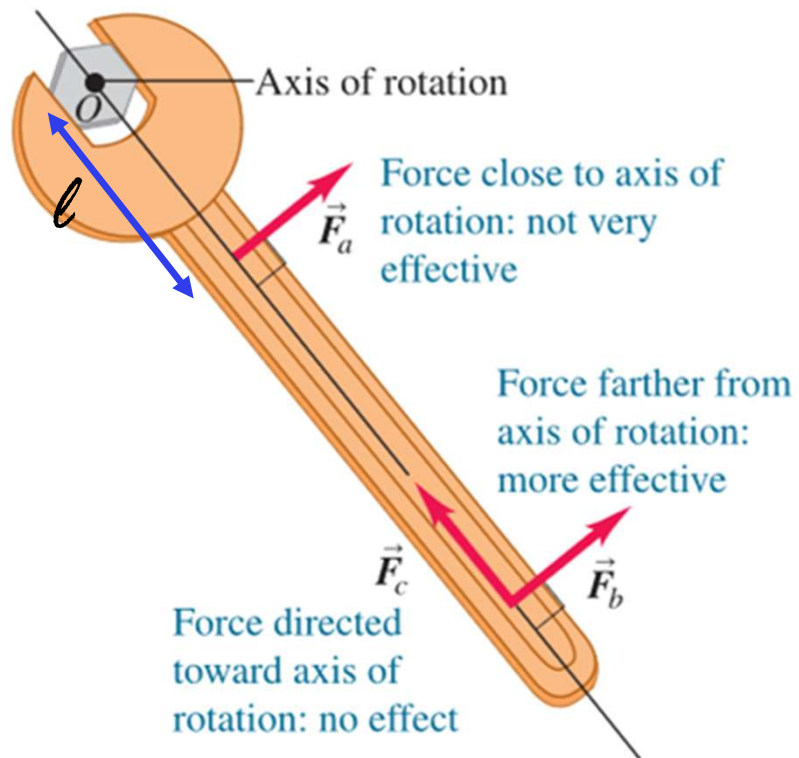
Rate of change of linear speed of that point

Rate of change of angular speed of body



Rotation – now with dynamic effects

Torque



For perpendicular forces ($\vec{F}_a$ and $\vec{F}_b$)

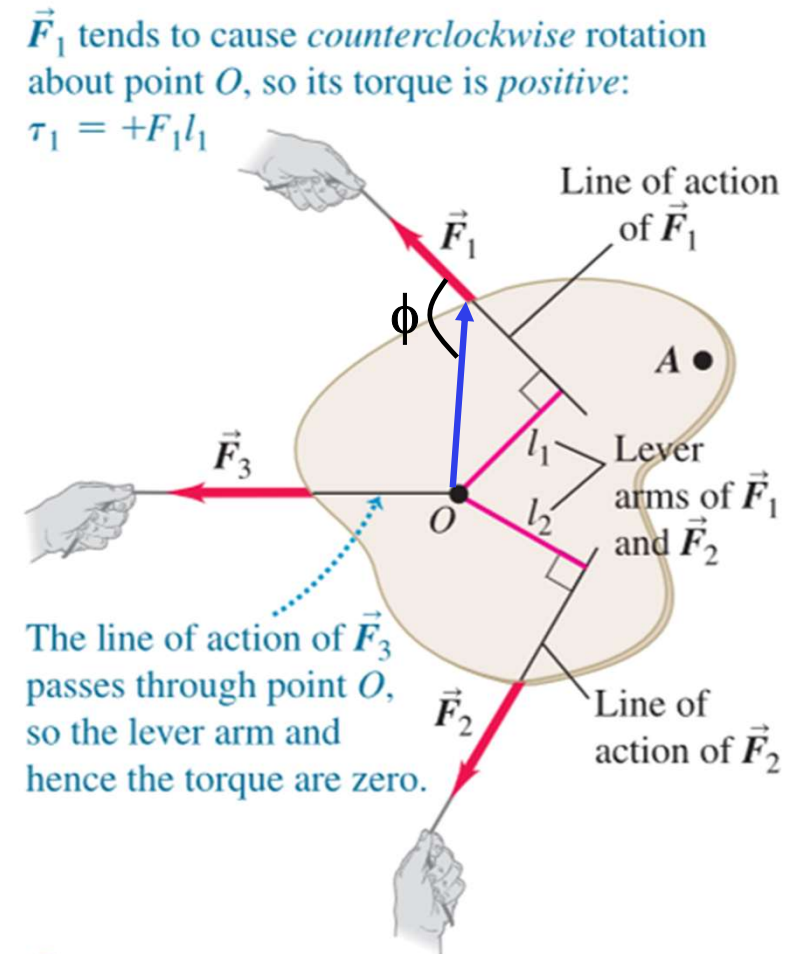
$$\tau = F \cdot \ell$$

Torque

Magnitude of torque due to force $\vec{F}$ relative to point O

$$\tau = Fl = rF \sin \phi = F_{\tan} r$$

Lever arm of $\vec{F}$
 Magnitude of $\vec{F}$
 Angle between $\vec{r}$ and $\vec{F}$
 Magnitude of $\vec{r}$ (vector from O to where $\vec{F}$ acts)
 Tangential component of $\vec{F}$



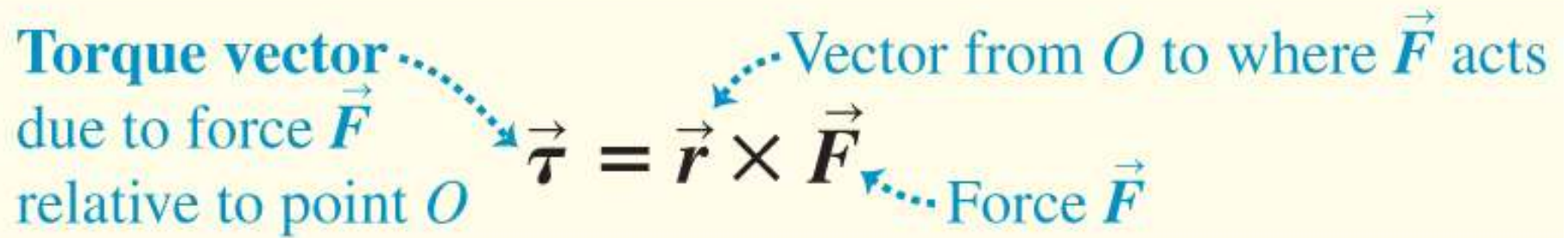
$\vec{F}_2$ tends to cause *clockwise* rotation about point O , so its torque is *negative*: $\tau_2 = -F_2 l_2$

Torque

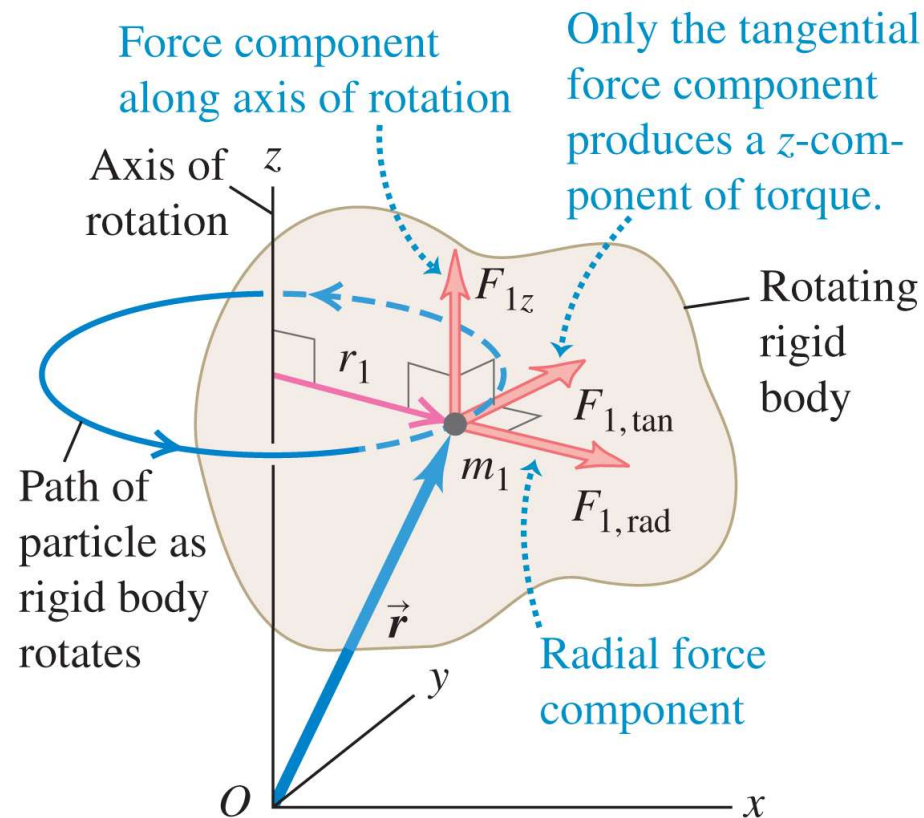
Torque vector due to force $\vec{F}$ relative to point O $\vec{\tau} = \vec{r} \times \vec{F}$

Vector from O to where $\vec{F}$ acts

Force $\vec{F}$



Forces on a rotating object



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Newton's 2nd law for rotation

Rotational analog of Newton's second law for a rigid body:

$$\sum \tau_z = I \alpha_z$$

Net torque on a rigid body about z-axis

Moment of inertia of rigid body about z-axis

Angular acceleration of rigid body about z-axis

Newton's second law: ... the object accelerates in the same direction as the net external force.

If there is a net external force on an object ...

$$\sum \vec{F} = m \vec{a}$$

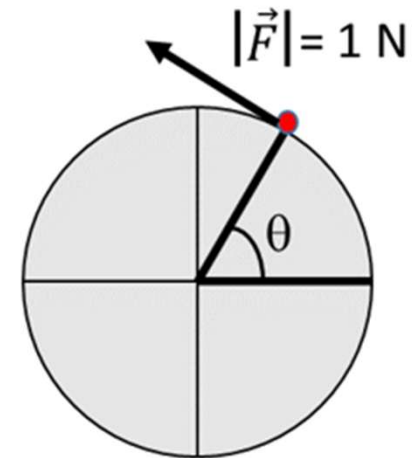
Mass of object

Problem from exam May 2022

Consider a homogeneous disk with mass $M = 1$ kg rotating through an axis passing perpendicular to it through its center of mass. At all times a tangential force to the edge of the disk, $\vec{F}$, with magnitude 1 N is applied at the red point (see Figure). Under these conditions the angular position of the red point is described by the equation $\theta(t) = 0.2 + 3t + t^2$, where t is given in seconds and θ is given in radians.

Data: Moment of inertia of a disk through an axis passing through its center of mass $I_{CM} = \frac{1}{2} M R^2$.

What is the radius of the disk?

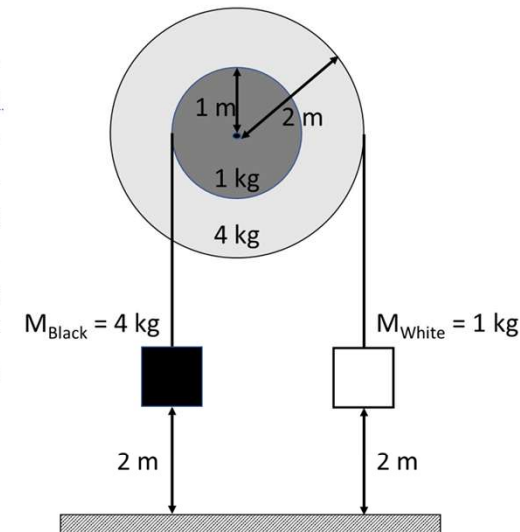


$$\theta(t) = 0.2 + 3t + t^2$$

Problem from exam August 2023

The system in the figure consists of two pulleys (homogeneous solid disks) stuck to each other. They are able to rotate around a fixed, horizontal axis that passes through the center of the disks. The small disk has a radius of $R_{\text{small}} = 1 \text{ m}$ and mass $M_{\text{small}} = 1 \text{ kg}$. The big disk has a radius of $R_{\text{big}} = 2 \text{ m}$ and mass $M_{\text{big}} = 4 \text{ kg}$. There is a black box with mass $M_{\text{Black}} = 4 \text{ kg}$ hanging from the small disk. The white box, with mass $M_{\text{White}} = 1 \text{ kg}$, is hanging from the big disk. Both boxes are 2 m above the floor. The system is freed from rest.

Which of the following statements is true?



- A) The white box will move down and the black box will move up.
- B) The white box will move up and the black box will move down.
- C) Just before the first box touches the floor the speed of the white box will be larger than that of the black box.
- D) Just before the first box touches the floor the speed of the white box will be smaller than that of the black box.
- E) Just before the first box touches the floor the speed of the white box will be the same as that of the black box.
- F) The boxes never move.
- G) Do not know.

Work under rotation

Work done by a torque τ_z

$$W = \int_{\theta_1}^{\theta_2} \tau_z d\theta$$

Upper limit = final angular position
Lower limit = initial angular position
Integral of the torque with respect to angle

Work done on a particle by a varying x -component of force F_x during straight-line displacement along x -axis

$$W = \int_{x_1}^{x_2} F_x dx$$

Upper limit = final position
Lower limit = initial position
Integral of x -component of force

Work done by a constant torque τ_z

$$W = \tau_z (\theta_2 - \theta_1) = \tau_z \Delta\theta$$

Torque
Final minus initial angular position = angular displacement

$$W = \int_{x_1}^{x_2} F_x dx = F_x \int_{x_1}^{x_2} dx = F_x (x_2 - x_1) \quad (\text{constant force})$$

Work under rotation

Total work done on a rotating rigid body = work done by the net external torque

Final rotational kinetic energy

$$W_{\text{tot}} = \int_{\omega_1}^{\omega_2} I\omega_z d\omega_z = \frac{1}{2}I\omega_2^2 - \frac{1}{2}I\omega_1^2$$

Initial rotational kinetic energy

(10.22)

$$W_{\text{tot}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 \quad (6.13)$$

Power due to a torque acting on a rigid body

$$P = \tau_z \omega_z$$

Torque with respect to rigid body's rotation axis

Angular velocity of rigid body about axis

(10.23)

Instantaneous power for a force doing work on a particle

$$P = \vec{F} \cdot \vec{v}$$

Force that acts on particle

Velocity of particle

(6.19)

Exercise

10.32 • An engine delivers 130 kW to an aircraft propeller at 2400 rev/min . (a) How much torque does the aircraft engine provide? (b) How much work does the engine do in one revolution of the propeller?



Angular momentum

Momentum of a particle
(a vector quantity)

$$\vec{p} = m\vec{v}$$

Particle mass
Particle velocity

(8.2)

Angular momentum of a rigid body rotating around a symmetry axis

$$\vec{L} = I\vec{\omega}$$

Moment of inertia of rigid body about symmetry axis
Angular velocity vector of rigid body

(10.28)

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Angular momentum of a particle relative to origin O of an inertial frame of reference

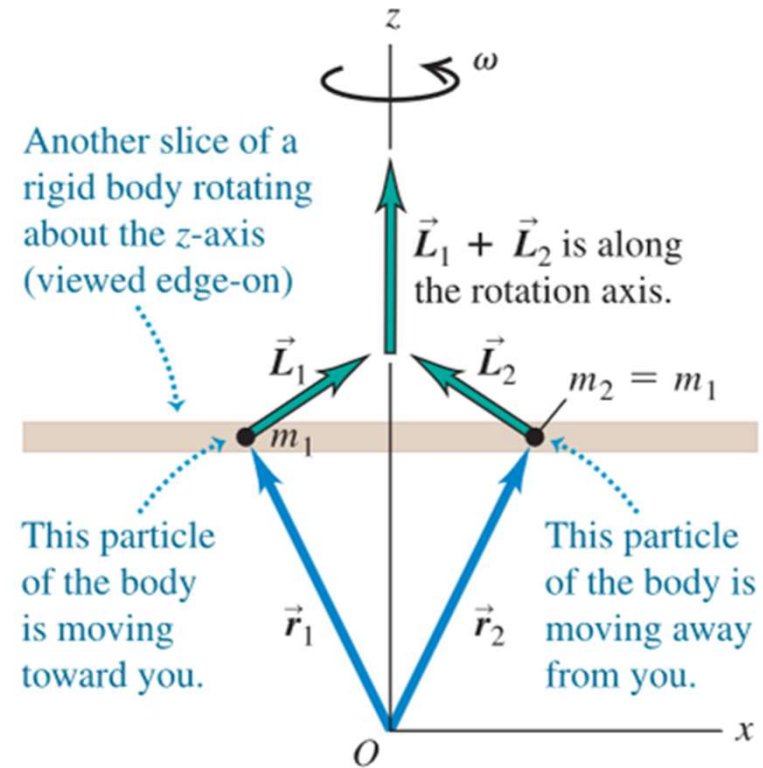
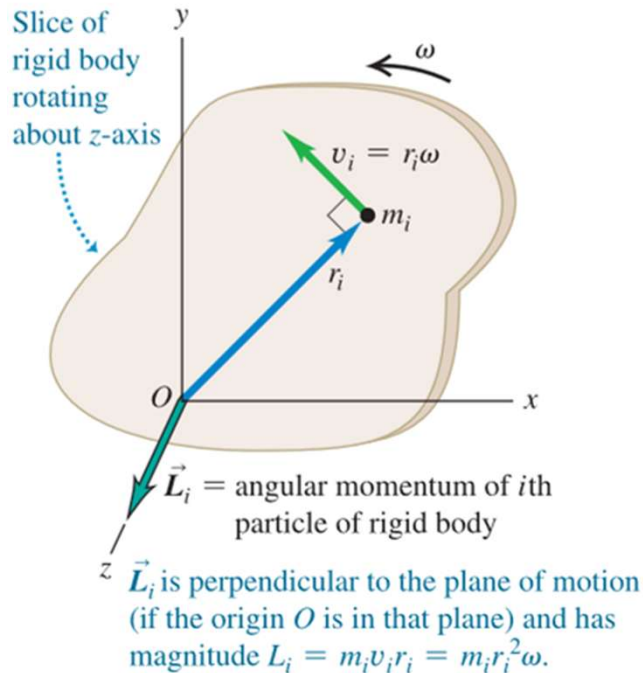
$$\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

Position vector of particle relative to O
Linear momentum of particle = mass times velocity

(10.24)

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Angular momentum



Angular momentum of a rigid body rotating around a symmetry axis

$$\vec{L} = I\vec{\omega}$$

Moment of inertia of rigid body about symmetry axis

Angular velocity vector of rigid body

(10.28)

Momentum theorem

**Newton's second law
in terms of momentum:**

The net external force
acting on a particle ...

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

... equals the rate of change
of the particle's momentum.

For a system of particles:

Sum of external torques
on the system

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

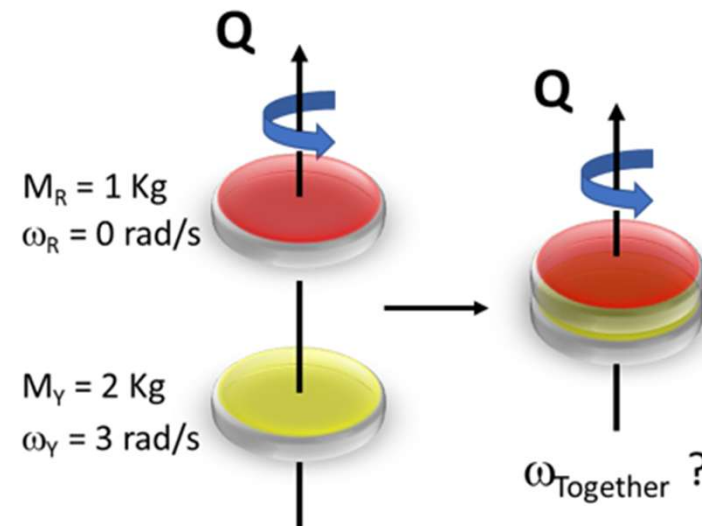
Rate of change of total
angular momentum $\vec{L}$
of system

CONSERVATION OF ANGULAR MOMENTUM When the net external torque acting on a system is zero, the total angular momentum of the system is constant (conserved).

Problem from exam May 2023

A yellow, solid disc with radius $R = 10$ cm and mass $M_Y = 2.0$ kg rotates at 3.0 rad/s through an axis Q that passes through its center of mass (see Figure). Above this yellow disc there is a red non-rotating, solid disc with the same radius and mass $M_R = 1.0$ kg. Shortly after the red disc is deposited on top of the yellow disc, both discs rotate with the same angular velocity.

What is the angular velocity of the discs once they are rotating together?



Summary of Linear-Rotary comparison

Straight-Line Motion with Constant Linear Acceleration

$$a_x = \text{constant}$$

$$v_x = v_{0x} + a_x t$$

$$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

$$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$$

Fixed-Axis Rotation with Constant Angular Acceleration

$$\alpha_z = \text{constant}$$

$$\omega_z = \omega_{0z} + \alpha_z t$$

$$\theta = \theta_0 + \omega_{0z} t + \frac{1}{2} \alpha_z t^2$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t$$

$$x \rightarrow \theta$$

$$v_x \rightarrow \omega$$

$$a_x \rightarrow \alpha$$

$$m \rightarrow I$$

Kinetic energy of a particle $\rightarrow K = \frac{1}{2}mv^2$

Mass of particle $\rightarrow$ Speed of particle

Rotational kinetic energy of a rigid body rotating around an axis $\rightarrow K = \frac{1}{2}I\omega^2$

Moment of inertia of body for given rotation axis $\rightarrow$ Angular speed of body

Summary of Linear-Rotary comparison

$\vec{F} \rightarrow \tau_z$ $\vec{p} \rightarrow \vec{L}$	Linear	Rotary
Newton's second law	$\sum \vec{F} = m\vec{a}$	$\sum \tau_z = I\alpha_z$
Work	$W = F_x(x_2 - x_1)$ (constant force)	$W = \tau_z(\theta_2 - \theta_1) = \tau_z \Delta\theta$
Power	$P = \vec{F} \cdot \vec{v}$	$P = \tau_z \omega_z$
Newton's second law (momentum)	$\sum \vec{F} = \frac{d\vec{p}}{dt}$	$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$